NANYANG TECHNOLOGICAL UNIVERSITY

MIDTERM II

MH2100 – Calculus III

October 2021 TIME ALLOWED: 90 minutes

Name:

Matric. no.: Tutor group:

INSTRUCTIONS TO CANDIDATES

1. **DO NOT TURN OVER PAPER UNTIL INSTRUCTED.**
2. This midterm paper contains **THREE (3)** questions.
3. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. **One** two-sided A4 piece of paper with handwritten notes is permitted.

QUESTION 1. **(16 marks)**

(i) [5 marks] Find the *maximum* and *minimum* values of $4x + 2y$ on the half disk given by $x^2 + y^2 \leq 16$ and $y \geq 0$. Also give the points (x, y) where the *maximum* and *minimum* values are achieved.

(The set of points (x, y) where $x^2 + y^2 \leq 16$ and $y \geq 0$ is a closed and bounded set so both the maximum and minimum must be achieved.)

(ii) [3 marks] Evaluate the iterated integral.

$$\int_{-1}^{1} \int_{0}^{1} xy \, \mathrm{d}x \, \mathrm{d}y.$$

(iii) [4 marks] By reversing the order of integration, evaluate the integral

$$\int_{0}^{30} \int_{y/6}^{5} e^{x^2} \, \mathrm{d}x \, \mathrm{d}y.$$

(iv) [4 marks] By reversing the order of integration, evaluate the integral

$$\int_{0}^{4} \int_{y/4}^{\sqrt[3]{y/4}} \sin(2x^2 - x^4) \, \mathrm{d}x \, \mathrm{d}y.$$

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QUESTION 2. **(17 marks)**

(i) [3 marks] Find the area of the region bounded by the lines $y = x^2$ and $y = 2x + 3$.

(ii) [2 marks] Find the Jacobian $\dfrac{\partial(x, y, z)}{\partial(u, v, w)}$ using the equations $x = -2u+5$, $y = 5v + 4$, and $z = 3w - 4$.

(iii) [4 marks] Use the transformation $u = x+y$ and $v = -2x+y$ to evaluate the integral

$$\iint_R 3x \, \mathrm{d}A,$$

where R is the parallelogram bounded by the lines

$y = -x + 1$, $y = -x + 4$, $y = 2x + 2$ and $y = 2x + 5$.

(iv) [4 marks] Use cylindrical coordinates to evaluate

$$\iiint_E x^2 y \, \mathrm{d}V,$$

where E is the solid that lies between the cylinders $x^2 + y^2 = 2$ and $x^2 + y^2 = 4$, above the xy-plane and below the plane $z = y$.

(v) [4 marks] Find the volume of the region bounded above by the sphere $x^2 + y^2 + z^2 = 9$ and below by the cone $z = \sqrt{x^2 + y^2}$.

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QUESTION 3. **(17 marks)**

(i) [4 marks] Evaluate the line integral

$$\int_C 3y \, \mathrm{d}s,$$

where C is the part of the curve $y = \dfrac{x^3}{3}$ where $0 \leq x \leq 2$.

(ii) [4 marks] Find the work done by the force field $\mathbf{F}(x, y) = (y - x)\mathbf{i} + x^2\mathbf{j}$ on a particle that moves along the curve given by $x = 3t$, $y = t^2 - 1$ from $(0, -1)$ to $(6, 3)$.

(iii) [3 marks] Evaluate the line integral

$$\int_C 8x^2yz \, \mathrm{d}x + 5z \, \mathrm{d}y - 4xy \, \mathrm{d}z,$$

where C has parametrisation $x = t$, $y = t^2$ and $z = t^3$, going from the point $(0, 0, 0)$ to $(1, 1, 1)$.

(iv) Let $\mathbf{F}(x, y, z) = (4z^2)\mathbf{i} + (\cos z)\mathbf{j} + (8xz - y \sin z + 1)\mathbf{k}$.

(1) [3 marks] Show that $\mathbf{F}$ is conservative by finding a potential function f such that $\mathbf{F} = \nabla f$.

(2) [3 marks] Evaluate the line integral $\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$ where C is the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ starting at the point with $t = 0$ and ending at the point with $t = \pi$.

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